Lecture 3

Subgroups, quotients and homogeneous spaces

Objectives. *A subgroup of a Lie group need not be a submanifold, and a coset space need not be a manifold. Closedness of the subgroup makes the coset space a manifold. Normality is a separate condition, needed for a group law. We test both conditions on an irrational line in a torus and on two descriptions of the sphere S^2.*

By the end of this lecture you will be able to:

- State the closed-subgroup theorem and the subgroup–subalgebra correspondence with their hypotheses.
- Prove that an irrational line in the two-torus is dense and not embedded.
- Identify a homogeneous space with a coset space by computing a stabiliser.
- Compute tangent spaces and isotropy actions for $\mathrm{SO}(3)/\mathrm{SO}(2)$ and $\mathrm{SU}(2)/\mathrm{U}(1)$.

3.1 Closed and immersed subgroups

A subset S of a manifold M carries the subspace topology of Lecture 1: its open sets are the sets $U \cap S$ with U open in M. The subset S is *closed* if $M \setminus S$ is open. In a manifold, S is closed exactly when S contains the limit of each convergent sequence of its points.

The *closure* $\overline{S}$ consists of the points whose every open neighbourhood meets S. It is the smallest closed set containing S. The set S is *dense* if $\overline{S} = M$. By the definition of closure, S is dense exactly when S meets every nonempty open subset of M.

In a manifold, every point of $\overline{S}$ is a limit of a sequence in S. To obtain the sequence, choose points of S in chart balls of radii tending to zero.

We also need compactness, in the form of finite subcovers. Lecture 5 uses this form to prove strict positivity of the Haar integral.

Definition 3.1 (Compact space)**.** A topological space is *compact* if every cover by open sets has a finite subcover.

A continuous image of a compact space is compact. Indeed, the inverse images of an open cover of the image form an open cover of the space. A finite subfamily of them covers the space, so the corresponding open sets cover the image.

A subset of $\mathbb{R}^n$ is compact exactly when it is closed and bounded (Heine–Borel). For subsets of $\mathbb{R}^n$ this agrees with the definition by sequences in *Linear Algebra for Physics*. For a set of matrices, closed and bounded must be read in the matrix space. For example, the set $\{tI : 0 < t \leq 1\}$ is bounded and it is closed in $\mathrm{GL}_n(\mathbb{R})$. It is not compact, because its closure in the matrix space contains the zero matrix.

Definition 3.2 (Immersions and embeddings)**.** A smooth map $f\colon N \to M$ is an *immersion* if df_p is injective for every $p \in N$. It is an *embedding* if it is an injective immersion and a homeomorphism from N onto $f(N)$ with the subspace topology.

The regular level sets of Lecture 1 are images of embeddings. An injective immersion can fail to be an embedding. The reason is that the topology carried over from N to $f(N)$ can be finer than the subspace topology, that is, it can have more open sets.

Definition 3.3 (Immersed and embedded subgroups)**.** Let G be a Lie group. An *immersed subgroup* of G is a subgroup H with a Lie group structure for which the inclusion $\iota\colon H \to G$ is an immersion. It is an *embedded subgroup* if ι is an embedding.

The differential $d\iota_e$ identifies $\mathrm{Lie}(H)$ with a subspace $\mathfrak{h} \subseteq \mathfrak{g}$. By Lecture 2, the differential of a Lie group homomorphism preserves brackets. Hence $\mathfrak{h}$ is a real Lie subalgebra of $\mathfrak{g}$.

Theorem 3.4 (Closed-subgroup theorem; cited)**.** *Let G be a Lie group and let $H \subseteq G$ be a subgroup that is a closed subset. Then H is an embedded submanifold of G, and with this structure H is a Lie group. Its Lie algebra is*

$$\mathfrak{h} = \{X \in \mathfrak{g} : \exp(tX) \in H \text{ for all } t \in \mathbb{R}\}.$$

The hypothesis concerns only the subset H. Smoothness of H is a conclusion. For example, $\mathrm{U}(n)$ and $\mathrm{SU}(n)$ are closed in $\mathrm{GL}_n(\mathbb{C})$, because they are defined by continuous equations.

A related regularity theorem removes a hypothesis from homomorphisms.

Theorem 3.5 (Automatic smoothness; cited)**.** *Every continuous homomorphism between finite-dimensional real Lie groups is smooth.*

See Bröcker–tom Dieck, Chapter I, Section 3, Proposition (3.12). In particular, finite-dimensional continuous representations have differentials. We use this fact from Lecture 4 onwards. In particular, by the lemma on smooth one-parameter subgroups in Lecture 2, every continuous one-parameter subgroup of a Lie group has the form $t \mapsto \exp(tX)$.

Theorem 3.6 (Subgroups and subalgebras; cited)**.** *Let G be a Lie group and let $\mathfrak{h} \subseteq \mathfrak{g}$ be a real Lie subalgebra. There is exactly one connected immersed subgroup $H \subseteq G$ with $\mathrm{Lie}(H) = \mathfrak{h}$. As a set, H is the subgroup generated by $\exp(\mathfrak{h})$. Every connected immersed subgroup arises in this way from its own Lie algebra.*

Warner, Chapter 3, proves Theorems 3.4 and 3.6 for general Lie groups. Hall, Section 5.9, proves Theorem 3.6 for matrix Lie groups. Kirillov Jr., Chapters 2 and 3, states these two subgroup theorems.

The two theorems run in opposite directions. The first starts with a subset and concludes that it is a manifold. The second starts with a subalgebra and constructs a group. That group need not be closed or embedded, so the word “immersed” in Theorem 3.6 cannot be replaced by “embedded”. The next section gives an example of a connected immersed

subgroup that is neither closed nor embedded.

3.2 An irrational line in the two-torus

The two-torus $T^2 = \mathrm{U}(1) \times \mathrm{U}(1)$ is a compact abelian Lie group. Identify its Lie algebra $i\mathbb{R} \oplus i\mathbb{R}$ with $\mathbb{R}^2$ by $(2\pi i a, 2\pi i b) \mapsto (a, b)$. The exponential map then becomes the surjective homomorphism

$$E\colon \mathbb{R}^2 \longrightarrow T^2, \qquad E(a,b) = (e^{2\pi i a}, e^{2\pi i b}),$$

with kernel $\mathbb{Z}^2$. For a real number α, the subalgebra $\mathfrak{h}_\alpha = \mathbb{R}(1,\alpha)$ gives the homomorphism and image

$$\varphi_\alpha(t) = E(t, \alpha t) = (e^{2\pi i t}, e^{2\pi i \alpha t}), \qquad L_\alpha = \varphi_\alpha(\mathbb{R}).$$

The velocity of φ_α at t is the left translate of $(2\pi i, 2\pi i\alpha) \neq 0$, so φ_α is an immersion. For rational α the image is a closed circle subgroup (Exercise Sheet 3). Irrational α behaves differently.

We look at the line only where it crosses one vertical circle. At times $x + k$ with k an integer, the first coordinate stays at $e^{2\pi i x}$, and the second moves by α modulo one at each step. Figure 3.1 shows this. So density reduces to one claim: the multiples of α, taken modulo one, come close to every number.

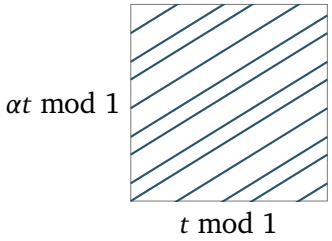


Figure 3.1: The line L_α for $0 \le t \le 8$ and $\alpha = (\sqrt{5} - 1)/2$, drawn in the unit square with opposite sides identified. Each new unit-time piece is shifted upwards by α modulo one.

Lemma 3.7 (Small multiples). *Let α be irrational and $n \ge 1$ an integer. There are integers q and m with $1 \le q \le n$ and*

$$0 < |q\alpha - m| < \frac{1}{n}.$$

Proof. For $j = 0, 1, \ldots, n$, put $f_j = j\alpha - \lfloor j\alpha \rfloor \in [0,1)$. These $n+1$ numbers lie in the n intervals $[s/n, (s+1)/n)$. By the pigeonhole principle, two of them, f_{j_1} and f_{j_2} with $j_1 < j_2$, lie in one interval. Set $q = j_2 - j_1$ and $m = \lfloor j_2\alpha \rfloor - \lfloor j_1\alpha \rfloor$. Then $q\alpha - m = f_{j_2} - f_{j_1}$, which has absolute value less than $1/n$. It is nonzero, since otherwise $\alpha = m/q$ would be rational. ∎

Proposition 3.8 (The irrational line). *Let α be irrational. Then:*

(a) φ_α is injective;
(b) L_α is dense in T^2;
(c) $L_\alpha \neq T^2$, so L_α is not closed;
(d) φ_α is not a homeomorphism onto L_α with the subspace topology.

Proof. (a) If $\varphi_\alpha(t) = \varphi_\alpha(s)$, then $t - s$ and $\alpha(t - s)$ are integers. If $t \neq s$, then α is the ratio of these two integers, so it is rational. Hence $t = s$.

(b) Fix a point $(e^{2\pi i x}, e^{2\pi i y})$ and an integer $n \geq 1$. First, take q and m from the small-multiples lemma, Lemma 3.7, and put $\delta = q\alpha - m \neq 0$. Let k be an integer nearest to $(y - \alpha x)/\delta$. Multiplying by $|\delta|$,

$$|k\delta - (y - \alpha x)| \leq \tfrac{1}{2}|\delta|.$$

Next, put $t = x + kq$. Here kq is an integer, and $\alpha kq = k\delta + km$ with km an integer. Therefore

$$\varphi_\alpha(t) = \left(e^{2\pi i x},\, e^{2\pi i(\alpha x + k\delta)}\right).$$

Finally, compare coordinates. The first coordinates of $\varphi_\alpha(t)$ and the chosen point agree. A chord of the unit circle is no longer than its arc, so $|e^{2\pi i u} - e^{2\pi i v}| \leq 2\pi|u - v|$ for all $u \in \mathbb{R}$ and $v \in \mathbb{R}$. The second coordinates therefore differ by at most $\pi|\delta| < \pi/n$. As n is arbitrary, L_α meets every neighbourhood of the chosen point. Since the chosen point is arbitrary, L_α is dense.

(c) Suppose $\varphi_\alpha(t) = (1, -1)$. Then t is an integer r, and $r\alpha = m + \frac{1}{2}$ for an integer m. Here $r \neq 0$, so $\alpha = (2m + 1)/(2r)$ would be rational. Thus $(1, -1) \notin L_\alpha$. A dense closed subset is the whole space, so L_α is not closed.

(d) For each n, take q_n and m_n from the small-multiples lemma, Lemma 3.7. Then

$$\varphi_\alpha(q_n) = \left(1, e^{2\pi i(q_n\alpha - m_n)}\right) \longrightarrow (1, 1) = \varphi_\alpha(0),$$

but $q_n \geq 1$ for every n. If the inverse map $L_\alpha \to \mathbb{R}$ were continuous at $(1, 1)$, then q_n would converge to 0. This is impossible. Hence the inverse map is not continuous at $(1, 1)$, and φ_α is not a homeomorphism onto L_α. ∎

So points close together in the torus can be far apart along the line. Figure 3.2 shows this near the identity. This is why the subgroup is not embedded.

Give L_α the Lie group structure transported from $\mathbb{R}$ by the bijection φ_α. It is then the connected immersed subgroup with Lie algebra $\mathfrak{h}_\alpha$ given by the subgroup–subalgebra correspondence, Theorem 3.6. By the irrational-line proposition, Proposition 3.8, the map φ_α is not a homeomorphism onto L_α with the subspace topology, so this subgroup is not embedded. The same proposition shows that L_α is not closed, so the closed-subgroup theorem, Theorem 3.4, does not apply to it. Since L_α is dense by the same proposition, its closure is the closed subgroup T^2. The Lie algebra $\mathbb{R}^2$ of T^2 is strictly larger than $\mathfrak{h}_\alpha$.

3.3 Coset spaces, actions and quotient groups

Let H be a subgroup of a Lie group G. The *coset space* $G/H = \{gH : g \in G\}$ has the quotient map π given by $\pi(g) = gH$. Its *quotient topology* declares $V \subseteq G/H$ open when $\pi^{-1}(V)$ is open in G. The group G acts on G/H by left translation, $\ell_g(xH) = gxH$.

Closedness is necessary for G/H to be Hausdorff. In a Hausdorff space each point is closed, because every other point has an open neighbourhood avoiding it. So if G/H is Hausdorff, then $H = \pi^{-1}(eH)$ is closed. For irrational α, the space T^2/L_α is therefore not a manifold.

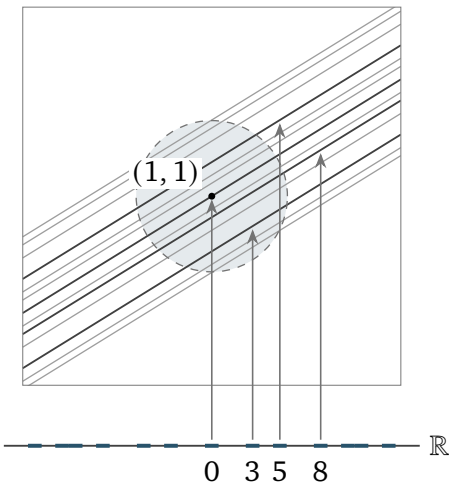


Figure 3.2: The square $[-\frac{1}{2}, \frac{1}{2}]^2$ is T^2, with opposite sides identified and the identity $(1,1)$ at the centre. Here $\alpha = (\sqrt{5}-1)/2$. The strands are the pieces of L_α for t in the heavy unit intervals around integers k of the line below. They are all the pieces with $|k| \le 13$ that meet the small dashed disc. The arrows show where the strands for $k = 0, 3, 5, 8$ come from.

Theorem 3.9 (Quotients by closed subgroups; cited)**.** *Let G be a Lie group and H a closed subgroup with Lie algebra $\mathfrak{h}$. Then:*

(1) G/H is Hausdorff and second countable. It has a unique smooth manifold structure for which π is a submersion*, that is, smooth with surjective differential at every point;*

(2) $\dim G/H = \dim G - \dim H$, and $d\pi_e$ induces an isomorphism $\mathfrak{g}/\mathfrak{h} \cong T_{eH}(G/H)$;

(3) the action $G \times G/H \to G/H$ is smooth;

(4) a map f from G/H to a manifold is smooth exactly when $f \circ \pi$ is smooth.

This theorem is in Bröcker–tom Dieck, Chapter I, Section 4, and in Warner, Chapter 3.

Definition 3.10 (Smooth actions)**.** A *smooth action* of G on a manifold M is a smooth map $G \times M \to M$, $(g,p) \mapsto g \cdot p$, with $e \cdot p = p$ and $g \cdot (g' \cdot p) = (gg') \cdot p$. The *stabiliser* of p is $G_p = \{g \in G : g \cdot p = p\}$, and the *orbit* of p is $G \cdot p$. The action is *transitive* if it has only one orbit. A manifold with a transitive smooth action is a *homogeneous space*.

Proposition 3.11 (Stabilisers and orbit maps)**.** *Let G act smoothly on M, fix $p \in M$, and put $H = G_p$.*

(a) H is a closed subgroup, and $G_{g\cdot p} = gHg^{-1}$.

(b) The orbit map $\beta(gH) = g \cdot p$ is a well-defined, injective, smooth map $G/H \to M$ with image $G \cdot p$. It is equivariant: $\beta(\ell_{g'}(gH)) = g' \cdot \beta(gH)$.

Proof. (a) Since $e \cdot p = p$, we have $e \in H$. If $g, g' \in H$, then $(gg') \cdot p = g \cdot (g' \cdot p) = p$ and $g^{-1} \cdot p = g^{-1} \cdot (g \cdot p) = p$. The point p is closed, because M is Hausdorff. So H is closed, as the preimage of p under the continuous map $g \mapsto g \cdot p$. Finally, $x \cdot (g \cdot p) = g \cdot p$ exactly when $(g^{-1}xg) \cdot p = p$, that is, when $x \in gHg^{-1}$.

(b) We have $g \cdot p = g' \cdot p$ exactly when $g^{-1}g' \in H$, that is, when $gH = g'H$. One direction gives

well-definedness, the other injectivity. Since H is closed by part (a), the quotient theorem, Theorem 3.9, applies to G/H. The composite $\beta \circ \pi(g) = g \cdot p$ is smooth, so β is smooth by the smoothness criterion of Theorem 3.9(4). Equivariance is the identity $(g'g) \cdot p = g' \cdot (g \cdot p)$. ∎

Theorem 3.12 (Homogeneous spaces; cited). *Let a Lie group G act smoothly and transitively on a manifold M, and let $p \in M$. Then the orbit map $\beta : G/G_p \to M$ is a G-equivariant diffeomorphism.*

The proof uses the second countability of G, which is part of the definition of a manifold in Lecture 1; see Warner, Chapter 3, or Kirillov Jr., Chapter 2.

We now compute T_pM from $\mathfrak{g}$. An element X of $\mathfrak{g}$ moves p with a velocity, namely the velocity of $t \mapsto \exp(tX) \cdot p$ at 0. For a transitive action, the elements with zero velocity form $\mathfrak{h}$, and the velocities fill T_pM. So $\mathfrak{g}/\mathfrak{h} \cong T_pM$. Next, the stabiliser H fixes p, so it acts on T_pM. This action tells the two sphere models below apart.

Two derivatives of the orbit map are used below. First, for $X \in \mathfrak{g}$, differentiate $\beta(\pi(\exp tX)) = \exp(tX) \cdot p$ at $t = 0$. This shows that $d\beta_{eH}(d\pi_e X)$ is the velocity of $t \mapsto \exp(tX) \cdot p$. This holds for every smooth action.

Second, assume now that the action is transitive. By the homogeneous-space theorem, Theorem 3.12, $d\beta_{eH}$ is an isomorphism $T_{eH}(G/H) \to T_pM$. Combining this with the tangent-space isomorphism of Theorem 3.9(2), we see that the composite $d\beta_{eH} \circ d\pi_e$ has kernel $\mathfrak{h}$ and induces an isomorphism $\mathfrak{g}/\mathfrak{h} \cong T_pM$.

As before, $H = G_p$. Let $k \in H$. The translation ℓ_k fixes eH. Its differential $d(\ell_k)_{eH}$ is called the *isotropy action* of k. The map $a_k : M \to M$, given by $a_k(p') = k \cdot p'$ for $p' \in M$, fixes p. Equivariance of β says $\beta \circ \ell_k = a_k \circ \beta$. The chain rule at eH therefore gives

$$d\beta_{eH} \circ d(\ell_k)_{eH} = d(a_k)_p \circ d\beta_{eH}.$$

So, transported by the isomorphism $d\beta_{eH}$, the isotropy action of k is the differential of a_k at p.

A coset space is not automatically a group.

Proposition 3.13 (Normality). *The rule $(xH)(yH) = xyH$ is well defined on G/H exactly when H is normal. It then makes G/H a group.*

Proof. Suppose the rule is well defined. Let $k \in H$ and $y \in G$. The representatives e and k of the coset H give the same product with yH, so $kyH = yH$. Hence $y^{-1}ky \in H$, and H is normal. Conversely, if H is normal, then for $k, k' \in H$,

$$xk\,yk' = xy\,(y^{-1}ky)\,k' \in xyH.$$

The group axioms are inherited from G. ∎

Theorem 3.14 (Quotient groups; cited). *Let H be a closed normal subgroup of a Lie group G. With the structure of Theorem 3.9, G/H is a Lie group and π is a homomorphism. The subalgebra $\mathfrak{h}$ is an ideal, and $\mathrm{Lie}(G/H) \cong \mathfrak{g}/\mathfrak{h}$ as Lie algebras.*

This theorem is in Warner, Chapter 3. Closedness of H makes G/H a manifold, and normality of H makes G/H a group. Since T^2 is abelian, L_α is normal, so T^2/L_α is a group. For irrational

α it is not a manifold. The two models of the sphere below are manifolds without a coset group law.

3.4 The sphere as SO(3)/SO(2)

The group $\mathrm{SO}(3)$ acts smoothly on $S^2 = \{v \in \mathbb{R}^3 : |v| = 1\}$ by matrix multiplication, with base point e_3. Let $J_1, J_2, J_3 \in \mathfrak{so}(3)$ be the matrices of Exercise Sheet 1, so that $J_k v = e_k \times v$. The rotations about the third axis are

$$R_\theta = \exp(\theta J_3) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Proposition 3.15 (Stabiliser of the north pole)**.** *The action of* $\mathrm{SO}(3)$ *on* S^2 *is transitive. The stabiliser of* e_3 *is* $H = \{R_\theta : \theta \in \mathbb{R}\} \cong \mathrm{SO}(2)$.

Proof. Let $v \in S^2$. Extend v to an orthonormal basis (u, w, v) of $\mathbb{R}^3$. The matrix g with columns u, w, v is orthogonal and $ge_3 = v$. Replacing u by $-u$ if necessary, we get $\det g = 1$. Thus the action is transitive.

Now let $ge_3 = e_3$. The third column of g is e_3. The rows of g are also orthonormal, because $gg^{\mathsf{T}} = I$. The third row is a unit vector with third entry one, so its other entries vanish. Hence

$$g = \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}, \qquad A^{\mathsf{T}} A = I, \qquad \det A = \det g = 1.$$

The first column of A is a unit vector, so it equals $(\cos\theta, \sin\theta)$ for some θ. The second column is a unit vector orthogonal to the first, and $\det A = 1$ selects $(-\sin\theta, \cos\theta)$. So $g = R_\theta$. Conversely, $R_\theta e_3 = e_3$. ∎

By the homogeneous-space theorem, Theorem 3.12, the map $gH \mapsto ge_3$ is an equivariant diffeomorphism $\mathrm{SO}(3)/\mathrm{SO}(2) \to S^2$. The dimensions agree: $3 - 1 = 2$.

The sphere is the level set $|v|^2 = 1$, so $T_{e_3} S^2 = e_3^\perp$ by Lecture 1. By the first derivative computation of the previous section, $d\beta_{eH} \circ d\pi_e$ sends $X \in \mathfrak{so}(3)$ to the velocity of $\exp(tX)e_3$, which is Xe_3. On the basis,

$$J_1 e_3 = e_1 \times e_3 = -e_2, \qquad J_2 e_3 = e_2 \times e_3 = e_1, \qquad J_3 e_3 = 0.$$

So $X \mapsto Xe_3$ has kernel $\mathbb{R}J_3 = \mathrm{Lie}(H)$ and image $e_3^\perp$. It induces the isomorphism $\mathfrak{so}(3)/\mathbb{R}J_3 \cong T_{e_3} S^2$, which is $d\beta_{eH}$ composed with the tangent-space isomorphism of Theorem 3.9(2).

For the isotropy action, the map $a_{R_\theta}(v) = R_\theta v$ is the restriction to S^2 of a linear map of $\mathbb{R}^3$. Its differential at e_3 is therefore R_θ restricted to $e_3^\perp$. In the coordinates (x, y) for $xe_1 + ye_2$, this is the rotation of the tangent plane by θ. Only $R_\theta = I$ acts trivially.

The stabiliser H is not normal. Take the rotation by $\pi/2$ about e_1,

$$g_0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad g_0 e_3 = -e_2.$$

By the conjugation formula of Proposition 3.11(a), $g_0 H g_0^{-1}$ is the stabiliser of $-e_2$. But $R_{\pi/2}(-e_2) = e_1$. Hence $R_{\pi/2} \in H$ and $R_{\pi/2} \notin g_0 H g_0^{-1}$. By the normality proposition, Proposition 3.13, there is no coset multiplication on $\mathrm{SO}(3)/\mathrm{SO}(2)$.

3.5 The sphere as SU(2)/U(1)

Let V be the real vector space of traceless Hermitian 2×2 matrices. It has the coordinates

$$\xi(x,y,z) = \begin{pmatrix} z & x-iy \\ x+iy & -z \end{pmatrix}, \qquad -\det \xi(x,y,z) = x^2+y^2+z^2.$$

The group $\mathrm{SU}(2)$ acts on V by $U\cdot\xi = U\xi U^*$. The matrix $U\xi U^*$ is Hermitian. Since $U^* = U^{-1}$, it has the same trace and determinant as ξ. Thus $\mathrm{SU}(2)$ acts smoothly on $S = \{\xi \in V : \det\xi = -1\}$, which the coordinates identify with S^2. In physics, with the Pauli matrices of Lecture 1, one writes $\xi(x,y,z) = x\sigma_1 + y\sigma_2 + z\sigma_3$, and S is the Bloch sphere of a two-level system. The base point is $N = \xi(0,0,1) = \mathrm{diag}(1,-1)$. With $A = \mathrm{diag}(i,-i)$ from Lecture 1, put $k_\theta = \exp(\theta A) = \mathrm{diag}(e^{i\theta}, e^{-i\theta})$ and $T = \{k_\theta : \theta \in \mathbb{R}\} \cong \mathrm{U}(1)$.

Proposition 3.16 (Stabiliser in SU(2)). *The action of* $\mathrm{SU}(2)$ *on* S *is transitive, and the stabiliser of* N *is* T.

Proof. Let $\xi \in S$. Its characteristic polynomial is $s^2 - (\mathrm{tr}\,\xi)s + \det\xi = s^2 - 1$, so its eigenvalues are ± 1. By the spectral theorem, $\xi = WNW^*$ for a unitary W whose columns are unit eigenvectors of ξ for the eigenvalues 1 and -1. Since W is unitary, its determinant has modulus one. Choose $\lambda \in \mathbb{C}$ with $|\lambda| = 1$ and $\lambda^2 = \overline{\det W}$. Then $U = \lambda W$ has determinant one, and $UNU^* = |\lambda|^2 WNW^* = \xi$.

For the stabiliser, $UNU^* = N$ is equivalent to $UN = NU$. Write U with entries a, b, c, d. Then

$$UN = \begin{pmatrix} a & -b \\ c & -d \end{pmatrix}, \qquad NU = \begin{pmatrix} a & b \\ -c & -d \end{pmatrix}.$$

Equality forces $b = c = 0$. A diagonal unitary matrix has $|a| = 1$, and $ad = 1$ gives $d = \overline{a}$. Thus $U = k_\theta$ for some θ. Conversely, k_θ commutes with N. ∎

By the homogeneous-space theorem, Theorem 3.12, the map $UT \mapsto UNU^*$ is an equivariant diffeomorphism $\mathrm{SU}(2)/T \to S \cong S^2$. Again the dimensions are $3 - 1 = 2$.

The map $d\beta_{eT} \circ d\pi_e$ sends $X \in \mathfrak{su}(2)$ to the velocity of $\exp(tX)N\exp(tX)^*$ at $t = 0$. By the product rule and since $X^* = -X$, this velocity is $XN + NX^* = [X, N]$. Use the basis A, B, C of $\mathfrak{su}(2)$ from Lecture 1, where

$$B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

Direct multiplication gives $[A, N] = 0$ and

$$[B,N] = \begin{pmatrix} 0 & -2 \\ -2 & 0 \end{pmatrix} = \xi(-2,0,0), \qquad [C,N] = \begin{pmatrix} 0 & -2i \\ 2i & 0 \end{pmatrix} = \xi(0,2,0).$$

The kernel of $X \mapsto [X, N]$ is $\mathbb{R}A = \mathrm{Lie}(T)$. The image of this map is $\{\xi(x, y, 0)\}$. In the coordinates, S is the unit sphere and N is its north pole, so this image is $T_N S$. Thus $X \mapsto [X, N]$ induces the isomorphism $\mathfrak{su}(2)/\mathbb{R}A \cong T_N S$, which is $d\beta_{eT}$ composed with the tangent-space isomorphism of Theorem 3.9(2).

For the isotropy action, the map $a_{k_\theta}(\xi) = k_\theta \xi k_\theta^*$ is the restriction to S of a linear map of V. Its differential at N is the same conjugation, acting on $T_N S$:

$$k_\theta\, \xi(x, y, 0)\, k_\theta^* = \begin{pmatrix} 0 & e^{2i\theta}(x - iy) \\ e^{-2i\theta}(x + iy) & 0 \end{pmatrix}.$$

Thus $x + iy \mapsto e^{-2i\theta}(x + iy)$, a rotation by -2θ. It is trivial exactly when $\theta \in \pi\mathbb{Z}$, that is, when $k_\theta = \pm I$. In physics, the element $k_{-\varphi/2} = \exp(-i\varphi\sigma_3/2)$ therefore rotates the tangent plane by φ, which is the half angle of spin one half.

The two models have the same space, circle stabilisers and tangent dimension. They differ in the isotropy action. Run each stabiliser circle once. Both R_θ and k_θ do this as θ runs over $[0, 2\pi]$. In the coordinates (x, y) used above, the tangent plane then turns once for SO(3), and twice in the opposite sense for SU(2).

The sense of rotation depends on the choice of parametrisation, because replacing k_θ by $k_{-\theta}$ reverses it. The invariant statement is the kernel of the isotropy action, which is trivial for SO(3) and $\{\pm I\}$ for SU(2). Moreover, $-I$ acts trivially on all of V, since $(-I)\xi(-I)^* = \xi$ for every $\xi \in V$. Lecture 4 uses the element $-I$ to relate the two groups.

The subgroup T is not normal in SU(2) (Exercise 3.6(ii)), so SU(2)/U(1) has no coset group law.

Self-checks

1. For which real α is L_α closed in T^2?
2. Is $(gH)(g'H) = gg'H$ a well-defined product on SO(3)/SO(2)?
3. Why is the stabiliser in SU(2) of every point of S a compact subgroup conjugate to T?

Answers. (1) Exactly for rational α. Let $\alpha = p/q$ with integers p and $q \geq 1$. Since q and $\alpha q = p$ are integers, $\varphi_\alpha(q) = (1, 1)$. So φ_α has period q, and $L_\alpha = \varphi_\alpha([0, q])$. This is a continuous image of a compact interval, hence compact in $\mathbb{C}^2$. It is therefore closed in $\mathbb{C}^2$, and so closed in T^2. For irrational α, the subgroup L_α is not closed by the irrational-line proposition, Proposition 3.8(c). (2) No. The stabiliser SO(2) is not normal, as g_0 shows. By the normality proposition, Proposition 3.13, the product is therefore not well defined. (3) By transitivity each point is $U \cdot N$. Its stabiliser is UTU^{-1} by the conjugation formula of Proposition 3.11(a). This is the image of the compact circle T under the continuous map $k \mapsto UkU^{-1}$, so it is compact.

Summary and reading

A closed subgroup of a Lie group is an embedded subgroup. A real Lie subalgebra determines a connected immersed subgroup, which need not be closed or embedded. For example,

the irrational line is dense and not embedded. For closed H, the coset space G/H is a manifold whose tangent space at eH is isomorphic to $\mathfrak{g}/\mathfrak{h}$. If a Lie group G acts smoothly and transitively on a manifold M, then M is equivariantly diffeomorphic to G/G_p for each $p \in M$. Normality is a further condition, needed for G/H to be a group. The sphere is diffeomorphic to both $\mathrm{SO}(3)/\mathrm{SO}(2)$ and $\mathrm{SU}(2)/\mathrm{U}(1)$. The two models have different isotropy actions.

Read Bröcker–tom Dieck, Chapter I, Section 4; Hall, Section 5.9; and Kirillov Jr., Chapter 2. Warner, Chapter 3, is the reference for the cited closed-subgroup, correspondence, quotient and homogeneous-space theorems. Lecture 4 studies the element $-I$ of $\mathrm{SU}(2)$ through covering groups.